

Overall Project Idea

Humara project:

real-world image ko AI navigation environment me convert karta hai

Aur phir:

- multiple path planning algorithms run karta hai
- unka comparison karta hai
- best navigation behavior analyze karta hai

Matlab:

image → grid → AI navigation → comparison → visualization

START

Yahan se pura pipeline start hota hai.

User:

- map/floorplan image deta hai
- ya occupancy-style environment input deta hai

Example:

- office map
- warehouse map
- room layout
- robotics environment

2 Load Occupancy Grid Map (Image)

System input image load karta hai.

Ye image:

- grayscale
- floorplan
- binary map
- occupancy map
- real environment render

kuch bhi ho sakti hai.

3 Image Preprocessing

(Grayscale + Thresholding)

Yahan image ko clean kiya jaata hai.

Grayscale

Image ko:

single intensity channel

me convert karte hain.

Reason:

processing faster aur simpler ho jaati hai

Thresholding

Yahan:

- white/light region → free space
- dark region → obstacle

ban jaata hai.

Matlab:

walkable vs non-walkable areas identify hote hain

Generate Binary Grid

Binary occupancy grid generate hota hai.

Yahan:

1 = free cell

0 = obstacle

Ab image mathematically grid ban gayi.

Ye robotics aur AI navigation ka standard representation hota hai.

5 Select Start & Goal Points

User:

- manually click kar sakta hai
- ya auto-detection use kar sakta hai

Start Point

Agent ki initial position.

Goal Point

Destination.

Neural Network Classifier (Optional)

Ye humara AI enhancement module hai.

Neural network:

walkable vs obstacle regions learn karta hai

Purpose:

- noisy maps improve karna
 - traversable regions refine karna
 - binary grid cleaner banana
-

Predict Traversable Area

NN refined prediction deta hai:

kaunsi jagah traversable hai

Matlab:

AI environment ko better samajhne ki koshish karta hai.

7 Path Planning Algorithms

Yahan main AI comparison hota hai.

Same environment pe:

multiple algorithms run hote hain

Aur compare hote hain.

◆ BFS

Breadth First Search

Layer-by-layer search karta hai.

Feature

Shortest path in unweighted grid

Drawback

Bahut zyada nodes explore karta hai.

◆ DFS

Depth First Search

Ek direction me deep jaata hai.

Feature

Fast ho sakta hai.

Drawback

Optimal path guarantee nahi karta.

◆ A* Search

A*:

cost + heuristic

use karta hai.

Formula:

$$f(n) = g(n) + h(n)$$

where:

- $g(n)$ = actual path cost
 - $h(n)$ = estimated remaining distance
-

◆ Manhattan Heuristic

Grid-based movement.

4-direction navigation

Best for:

- up/down/left/right movement
-

◆ Euclidean Heuristic

Straight-line distance use karta hai.

More realistic geometric estimation.

◆ Chebyshev Heuristic

Diagonal movement environments ke liye useful.

◆ MDP

Markov Decision Process

Yahan environment:

state transition model

ban jaata hai.

MDP:

future expected rewards maximize

karta hai.

Using:

Value Iteration

◆ Q-Learning

Reinforcement Learning Agent

Agent:

- environment me move karta hai
 - rewards collect karta hai
 - best actions learn karta hai
-

ϵ -Greedy Exploration

Initially:

random exploration

Later:

best learned policy follow

Q-Values

Agent learn karta hai:

kis state me kaunsa action best hai

8 Metrics Collection

Har algorithm ke liye collect hota hai:

Path Length

Kitne cells travel hue.

Nodes Explored

Kitni search effort lagi.

Execution Time

Algorithm kitna fast tha.

9 Compare Algorithms

Yahan:

all algorithms side-by-side compare
hote hain.

Comparison basis:

- efficiency
 - speed
 - search effort
 - path quality
-

10 Visualization Results

Final paths:

map ke upar overlay
kiye jaate hain.

User visually dekh sakta hai:

- kaunsa path best hai
- kis algorithm ne kya route liya

Dashboard Features

Dashboard me:

- algorithm comparison table
- RL training curves
- neural network loss graph
- heuristic comparison
- heatmaps
- best algorithm analysis

sab show hota hai.

Final Goal

Humara system:

vision-based AI navigation framework

hai jo:

- image understanding
- path planning
- reinforcement learning
- neural refinement

sab combine karta hai.